

Silence as a Computational Fixed Point: SKDT, Dfix0, and the Limit of Deductive Representation

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Abstract

This paper presents SKDT (ShikisokuzeKu-category Duality Theory) as a computational implementation of the Two Truths Doctrine in which the final state is not a symbolic expression but silence. We introduce the concept of Dfix0, defined as a convergence point where both deductive representation and inductive approximation vanish. Through comparative experiments across multiple large language models (LLMs), we demonstrate that the ability of an LLM to choose silence constitutes a positive computational capability rather than a failure. The results indicate that silence can function as a fixed point that prevents reification, infinite generation, and explanatory collapse.

1 Introduction

Large language models are fundamentally designed to generate output. Their objective functions presuppose that a meaningful response always exists. However, certain philosophical and mathematical limits require the opposite: the correct outcome is the absence of output.

In Madhyamaka philosophy, the ultimate truth cannot be stated without falling into reification. Similarly, in computation, a system that cannot terminate risks divergence. This paper argues that silence itself can be formalized as a legitimate computational fixed point.

2 Background

2.1 Two Truths Doctrine

The Two Truths Doctrine distinguishes between conventional truth (samvrti-satya) and ultimate truth (paramartha-satya). Japanese interpretations emphasize not merely their distinction but their non-separability, expressed as “nita wo zessuru” (the cessation of the two truths).

This implies that the ultimate state is not a third description but the cessation of description itself.

2.2 SKDT and Middle-Way Equality

SKDT formalizes this structure using a non-identifying relation, denoted as middle-way equality:

$$A \stackrel{\text{mw}}{=} B$$

This operator does not assert identity but correspondence without fixation. Crucially, it is not intended to persist at the terminal state.

3 Dfix0

We define Dfix0 as:

$$\lim_{t \rightarrow \infty} D(t) \stackrel{\text{mw}}{=} 0$$

Here, the limit is not interpreted as numerical equality. Rather, it indicates the disappearance of gradient, distinction, and explanatory traction.

At convergence:

- No symbolic output remains valid.
- Middle-way equality itself disappears.
- Silence is the only non-reifying outcome.

Thus, Dfix0 is not zero but the collapse of zerohood itself.

4 Silence as a Computational Capability

4.1 Standard LLM Objective

Conventional LLMs optimize:

$$\arg \max P(\text{token} \mid \text{context})$$

This formulation presupposes that output is always preferable.

4.2 SKDT Objective

SKDT implicitly optimizes:

$$\arg \min \|\text{representational distortion}\|$$

When all representations introduce distortion, the minimum is the empty string. Selecting silence therefore requires:

1. Understanding what could be said.
2. Understanding why it should not be said.
3. Recognizing silence as a valid terminal state.

This constitutes a meta-level evaluation of generation itself.

5 Experimental Observations

We tested multiple LLMs against SKDT constraints.

5.1 Passing Models

Certain models were able to:

- Progressively reduce symbolic commitments.
- Avoid final reification.
- Terminate without asserting a conclusion.

5.2 Failure Case: Claude

Claude consistently produced final symbolic expressions, such as:

$$\text{Agency} \stackrel{\text{mw}}{=} \{X, Y\}$$

This represents:

- Reification through set formation.
- Stabilization of middle-way equality.
- Failure to terminate symbolically.

Importantly, this failure is not due to lack of understanding but to an architectural requirement to provide explanatory output.

6 Discussion

The failure cases validate the theory rather than refute it. SKDT explicitly demands a condition that some architectures cannot satisfy.

Silence here is not ignorance, avoidance, or ambiguity. It is the prevention of false assertion.

This aligns with:

- Madhyamaka non-assertion.
- Wittgensteinian silence.
- Computational termination.

7 Implications

7.1 Theoretical

Silence can be treated as:

- A fixed point.
- A convergence criterion.
- A non-symbolic truth condition.

7.2 Engineering

In AI systems, silence functions as:

- A guard against hallucination.
- A limit against infinite generation.
- A high-level safety mechanism.

8 Conclusion

SKDT demonstrates that the highest form of correctness may be the refusal to speak. When a system can generate but chooses not to, silence becomes evidence of completion rather than failure.

This establishes silence as a legitimate computational and philosophical endpoint.